

CREATE++: Extending Cross-Representation Alignment with Signed and Higher-Order Signals

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Abstract—Transformer architectures have become the dominant approach in sequential recommender systems due to their ability to capture dependencies within user interaction histories. However, these models often consider sequence elements in isolation, failing to fully leverage the global collaborative context inherent in user–item relationships. While recent frameworks like CREATE [1] address this by aligning sequential encoders with graph neural networks (GNNs), they remain largely limited to unsigned bipartite graphs that cannot model explicit negative feedback or higher-order multi-way relationships. To fill this gap, we propose two architectural extensions: CREATE-Pone and CREATE-Uni. By integrating signed graph modeling inspired by Pone-GNN [2] to disentangle interest from disinterest and hypergraph structures via the UniGNN [3] framework to capture complex higher-order correlations, our framework provides a more expressive and nuanced grounding for sequential models. This dual approach significantly enriches the structural knowledge transferred to the transformer while preserving the efficient asymmetric inference protocol required for production-scale recommendation. Code for the project is available at <https://github.com/JConquers/CREATE->.

Index Terms—Recommender Systems, Sequential Recommendation, Signed Graphs, Hypergraphs

I. INTRODUCTION

Modern recommender systems must reconcile two distinct perspectives of user behavior: short-range sequential dynamics and long-range collaborative structures. While Graph Neural Networks (GNNs) have emerged as the state-of-the-art for capturing global collaborative signals through recursive neighborhood aggregation, they are often less effective at modeling the strictly ordered, temporal dependencies inherent in user interaction histories. Conversely, Transformer-based architectures excel at capturing these fine-grained sequential patterns but struggle to encode the broader structural context of the global user–item interaction graph. To bridge this gap, the recently proposed CREATE framework [1] introduces a paradigm of cross-representation knowledge transfer. By using a GNN as a global structural grounding to enrich a transformer-based sequential encoder during training, CREATE achieves superior performance by combining the strengths of both architectures. This alignment allows the model to inherit global collaborative wisdom while maintaining an efficient asymmetric inference protocol that utilizes only the sequential encoder at test time.

Despite its effectiveness, the original CREATE framework relies on an unsigned bipartite graphs, which introduce two significant structural bottlenecks. First, it suffers from *polarity blindness*; by treating all observed interactions as positive, it

ignores explicit negative feedback (e.g., low ratings or skips), leading to label noise and an inability to model user aversion. Second, it is limited by *pairwise collapse*, where higher-order multi-way relationships—such as a user’s entire session history or co-purchase groups—are reduced to isolated binary links. This simplification results in a loss of complex collaborative context that pairwise edges cannot distinguish. Third, the framework is hindered by *exposure bias*. Because unsigned bipartite GNNs typically treat all unobserved interactions as negative signals, the model fails to distinguish between items a user actively disliked and those they simply never encountered, leading to suboptimal generalization.

In this paper, we propose two extensions to the CREATE framework to recover this lost structural information:

- **CREATE-Pone (Signed Variant):** We replace the unsigned encoder with a signed GNN [2] that learns dual interest and disinterest embeddings. The alignment framework is extended to a multi-objective task: the sequential user representation is aligned with the global interest view to maximize knowledge transfer, while simultaneously being constrained to be orthogonal to the disinterest view. This allows the transformer to inherit collaborative preferences while explicitly suppressing signals associated with user aversion.
- **CREATE-Uni (Hypergraph Variant):** We leverage a unified hypergraph framework [3] to model user histories as hyperedges. This transition enables the model to capture multi-way collaborative signals with 1-GWL expressive power, strictly exceeding the topological limits of traditional bipartite GNNs like LightGCN.

Our contributions include the formalization of these signed and higher-order alignment objectives, a theoretical analysis of their expressive advantages, and an extensive empirical evaluation across five benchmark datasets. The remainder of this paper is organized as follows: Section II reviews related work, Section III presents the methodology, Section IV provides theoretical analysis, and Section V describes the experimental setup.

II. RELATED WORK

The evolution of recommendation systems has moved from modeling isolated sequences to capturing global structural dependencies, and recently, toward the alignment of these two distinct paradigms.

A. Sequential Recommendation

Sequential recommendation focuses on predicting the next item based on a user's chronological interaction history. Early models utilized Recurrent Neural Networks (RNNs), but the field has been revolutionized by Transformer-based architectures. *SASRec* [4] adopted a self-attention mechanism to identify relevant items in a sequence, while *BERT4Rec* [5] introduced bidirectional modeling via masked language objectives. To improve robustness, *CLASRec* [6] integrated contrastive learning to ensure that different augmentations of the same sequence yield similar representations. However, these models primarily focus on local dependencies and often lack a global perspective of item-item collaborative signals.

B. Graph-based Collaborative Filtering

Graph Neural Networks (GNNs) explicitly model user-item interactions as a bipartite graph to capture long-range collaborative signals. This paradigm involves learning d -dimensional node embeddings for both users ($\mathbf{e}_u$) and items ($\mathbf{e}_i$) by aggregating information from their respective neighborhoods. *NGCF* [7] first introduced neighborhood aggregation to recommendation, but was later simplified by *LightGCN* [8], which removed non-linear activations and weight matrices to improve efficiency and performance:

$$\mathbf{e}_u^{(k+1)} = \sum_{i \in \mathcal{N}(u)} \frac{1}{\sqrt{|\mathcal{N}(u)| |\mathcal{N}(i)|}} \mathbf{e}_i^{(k)} \quad (1a)$$

$$\mathbf{e}_i^{(k+1)} = \sum_{u \in \mathcal{N}(i)} \frac{1}{\sqrt{|\mathcal{N}(i)| |\mathcal{N}(u)|}} \mathbf{e}_u^{(k)} \quad (1b)$$

After K layers of propagation, the final embeddings are typically a weighted sum or average of the embeddings at each layer. Predicted scores are then computed via the inner product between the final user and item embeddings: $\hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}_i$. While GNNs are highly effective for capturing global structural patterns, they suffer from two primary issues in dynamic settings. First, they are inherently limited by the 1-dimensional Weisfeiler-Leman (1-WL) test, meaning they primarily capture pairwise dependencies and fail to recognize higher-order multi-way correlations. Second, they face the folding-in problem: generating recommendations for a new user or a new session at inference time requires re-running expensive message-passing operations across the global graph or retraining the model entirely. GNNs typically lack the flexibility to handle the real-time evolution of user interests due to this high computational cost at test time. To avoid this bottleneck, modern hybrid systems utilize sequential models for inference, relying on the GNN only during the training phase to provide an enriched, structural grounding for the sequential item embeddings.

C. Cross-Representation Knowledge Transfer

The *CREATE* framework [1] serves as the primary basis for this work, bridging the gap between local sequential patterns and global collaborative structures through representation alignment. The framework utilizes a dual-encoder architecture:

a sequential encoder (such as *SASRec* [4] or *BERT4Rec* [5]) processes the interaction history to generate a temporal user representation $\mathbf{h}_u$, while a graph encoder (typically *LightGCN* [8] or *UltraGCN*) processes the global bipartite graph to produce structural user embeddings $\mathbf{z}_u$ and item embeddings $\mathbf{z}_i$. The core innovation of *CREATE* is the use of the *Barlow Twins* objective [9] to align these two distinct views. After batch standardization of $\mathbf{h}_u$ and $\mathbf{z}_u$, the model computes a cross-correlation matrix $\mathcal{C}$, where the alignment loss $\mathcal{L}_{align}$ forces the sequential representation to inherit structural context by driving $\mathcal{C}$ toward the identity matrix:

$$\mathcal{L}_{align} = \sum_i (1 - C_{ii})^2 + \lambda \sum_{i \neq j} C_{ij}^2 \quad (2)$$

The first term ensures invariance (alignment), while the second term minimizes redundancy between the dimensions. To ensure the sequential encoder effectively learns from the structural grounding, the framework adopts a joint training objective:

$$\mathcal{L} = \mathcal{L}_{local} + \omega_{global} \mathcal{L}_{global} + \omega_{align} \mathcal{L}_{align} \quad (3)$$

Here, $\mathcal{L}_{local}$ represents the next-item prediction loss for the sequential model, $\mathcal{L}_{global}$ is the recommendation loss for the GNN (e.g., BPR), and ω are weighting hyperparameters. Predicted scores are then computed via the inner product between the sequential user and graph item embeddings: $\hat{y}_{ui} = \mathbf{h}_u^\top \mathbf{z}_i$, thus the model eliminates the need for the graph encoder during inference, allowing for efficient real-time recommendation using only the sequential encoder. Our work extends this objective by replacing the standard bipartite GNN with more expressive signed and hypergraph encoders.

D. Hypergraph Modeling and UniGNN

To capture multi-way relationships (e.g., items co-purchased in a single session), hypergraph-based models have emerged. *HGNN* [10] and *HNHN* [11] provide frameworks for high-order correlation modeling. *UniGNN* [3] generalized these approaches into a two-stage message-passing framework:

$$\begin{cases} \mathbf{h}_e = \phi_1(\{\mathbf{x}_j\}_{j \in e}) & \text{(Node to Hyperedge)} \\ \tilde{\mathbf{x}}_i = \phi_2(\mathbf{x}_i, \{\mathbf{h}_e\}_{e \in E_i}) & \text{(Hyperedge to Node)} \end{cases} \quad (4)$$

Furthermore, *DHCN* [12] demonstrated that modeling sessions as hyperedges allows the capture of higher-order signals that traditional graphs collapse into pairwise links. Our *CREATE-Uni* extension leverages this 1-GWL expressive power to enrich sequential grounding.

E. Signed Graph Learning

Despite the success of alignment frameworks, most GNN-based recommenders operate on unsigned bipartite graphs, treating all observed interactions as positive signals. This introduces significant label noise and exposure bias, as the model cannot distinguish between items a user actively disliked and those they simply never encountered. To address this, *Pone-GNN* [2] introduces a signed graph framework that explicitly models negative feedback by partitioning the

global interaction set $\mathcal{E}$ into two disjoint subsets: a positive interaction graph $\mathcal{G}_p = (\mathcal{V}, \mathcal{E}_p)$ and a negative interaction graph $\mathcal{G}_n = (\mathcal{V}, \mathcal{E}_n)$. The framework employs two independent message-passing branches to learn dual representations for each node: an interest embedding $\mathbf{z} \in \mathbb{R}^d$ derived from $\mathcal{G}_p$, and a disinterest embedding $\mathbf{v} \in \mathbb{R}^d$ derived from $\mathcal{G}_n$. These node embeddings are optimized via a contrastive learning loss that ensures the latent spaces are well-separated and informative. For the final recommendation task, the model utilizes a dual feedback-aware loss $\mathcal{L}_{DF}$ to maximize the likelihood of observed interactions within their respective polarities:

$$\mathcal{L}_{DF} = - \sum_{(u,i) \in \mathcal{E}_p} \log \sigma(\mathbf{z}_u^\top \mathbf{z}_i) - \sum_{(u,j) \in \mathcal{E}_n} \log \sigma(-\mathbf{v}_u^\top \mathbf{v}_j). \quad (5)$$

Our *CREATE-Pone* extension integrates this signed modeling into the alignment framework, enabling the sequential encoder to explicitly suppress aversion signals.

III. PROPOSED METHODOLOGY

Let $\mathcal{U}$ and $\mathcal{I}$ denote the sets of users and items, respectively. Each user $u \in \mathcal{U}$ has a timestamped interaction history $H_u = (i_1^u, \dots, i_{|H_u|}^u)$. We retain the shared embedding layer, the sequential encoder (SASRec with full softmax cross-entropy loss L_{local}), and the two-phase training procedure (graph/hypergraph warm-up followed by joint optimization) from the original CREATE framework.

A. CREATE-Pone: Signed Graph Variant

We partition $\mathcal{E}$ into positive interactions $\mathcal{E}_p$ (ratings ≥ 4 or explicit likes) and negative interactions $\mathcal{E}_n$ (ratings ≤ 3 or explicit dislikes), yielding two disjoint graphs. This fixed threshold is a practical initialization; recent work such as ReFiNe and ReFiNe++ [13], [14] suggests LLM-assisted refinement can reclassify ambiguous low ratings by modeling user-specific preference patterns. Exploring these adaptive heuristics is a promising future direction:

$$\mathcal{G}_p = (\mathcal{U}, \mathcal{I}, \mathcal{E}_p), \quad \mathcal{G}_n = (\mathcal{U}, \mathcal{I}, \mathcal{E}_n).$$

Each node $v \in \mathcal{U} \cup \mathcal{I}$ is assigned two initial embeddings:

$$\mathbf{z}_v^{(0)}, \mathbf{v}_v^{(0)} \in \mathbb{R}^d$$

(interest from $\mathcal{G}_p$, disinterest from $\mathcal{G}_n$).

We construct two training sets for mini-batch learning:

$$D_p = \{(u, i, j) \mid (u, i) \in \mathcal{E}_p, j \notin \mathcal{N}_p(u), u \in \mathcal{U}, i, j \in \mathcal{I}\}, \quad (6a)$$

$$D_n = \{(u, i, j) \mid (u, i) \in \mathcal{E}_n, j \notin \mathcal{N}_n(u), u \in \mathcal{U}, i, j \in \mathcal{I}\}. \quad (6b)$$

Mini-batches sampled from these sets are denoted by $B_p \subset D_p$ and $B_n \subset D_n$, respectively.

For layers $k = 0, \dots, K-1$, we perform dual-branch propagation on both signed graphs.

On the positive graph $\mathcal{G}_p$ and negative graph $\mathcal{G}_n$, updates are

$$\mathbf{z}_u^{(k+1)} = \frac{1 + \epsilon_p^{(k)}}{|\mathcal{N}_p(u)|} \mathbf{z}_u^{(k)} + \sum_{i \in \mathcal{N}_p(u)} \frac{\mathbf{z}_i^{(k)}}{\sqrt{|\mathcal{N}_p(u)| |\mathcal{N}_p(i)|}}, \quad (7a)$$

$$\mathbf{z}_i^{(k+1)} = \frac{1 + \epsilon_p^{(k)}}{|\mathcal{N}_p(i)|} \mathbf{z}_i^{(k)} + \sum_{u \in \mathcal{N}_p(i)} \frac{\mathbf{z}_u^{(k)}}{\sqrt{|\mathcal{N}_p(i)| |\mathcal{N}_p(u)|}}, \quad (7b)$$

$$\mathbf{v}_u^{(k+1)} = \frac{1 + \epsilon_n^{(k)}}{|\mathcal{N}_n(u)|} \mathbf{v}_u^{(k)} + \sum_{i \in \mathcal{N}_n(u)} \frac{\mathbf{v}_i^{(k)}}{\sqrt{|\mathcal{N}_n(u)| |\mathcal{N}_n(i)|}}, \quad (7c)$$

$$\mathbf{v}_i^{(k+1)} = \frac{1 + \epsilon_n^{(k)}}{|\mathcal{N}_n(i)|} \mathbf{v}_i^{(k)} + \sum_{u \in \mathcal{N}_n(i)} \frac{\mathbf{v}_u^{(k)}}{\sqrt{|\mathcal{N}_n(i)| |\mathcal{N}_n(u)|}}. \quad (7d)$$

In Eq. (7a) and (7b) are computed on the positive graph $\mathcal{G}_p$, while (7c) and (7d) are computed analogously on the negative graph $\mathcal{G}_n$. The final embeddings are layer-averaged:

$$\mathbf{z}_v = \frac{1}{K} \sum_{k=0}^{K-1} \mathbf{z}_v^{(k)}, \quad \mathbf{v}_v = \frac{1}{K} \sum_{k=0}^{K-1} \mathbf{v}_v^{(k)}. \quad (8)$$

The global loss is defined as follows.

- Dual feedback-aware loss L_{DF} :

$$L_{DF} = - \sum_{(u,i,j) \in B_p} \ln \sigma(\hat{y}_{u,i} - \hat{y}_{u,j}) - \sum_{(u,i,j) \in B_n} \ln \sigma(\hat{y}_{u,j} - \hat{y}_{u,i}), \quad (9)$$

with

$$\hat{y}_{u,i} = \begin{cases} \mathbf{z}_u^\top \mathbf{z}_i, & \text{if } (u, i, j) \in B_p, \\ b \cdot \mathbf{v}_u^\top \mathbf{v}_i, & \text{if } (u, i, j) \in B_n, \end{cases} \quad (10)$$

$$\hat{y}_{u,j} = \begin{cases} \mathbf{z}_u^\top \mathbf{z}_j, & \text{if } (u, i, j) \in B_p, \\ \mathbf{v}_u^\top \mathbf{v}_j, & \text{if } (u, i, j) \in B_n. \end{cases}$$

Here, $\hat{y}_{u,i}$ denotes the predicted similarity score between user u and item i .

- Contrastive learning loss L_{CL} :

$$L_{CL} = -\frac{1}{N} \sum_{\substack{(u,i,j) \in B_p \\ (u,i',j') \in B_n}} \log \frac{\exp(\mathbf{z}_u^\top \mathbf{z}_i / \tau)}{\exp(\mathbf{z}_u^\top \mathbf{z}_i / \tau) + \exp(\mathbf{v}_u^\top \mathbf{v}_{i'} / \tau)}. \quad (11)$$

The total global loss is

$$L_{\text{global}} = L_{DF} + L_{CL} \quad (12)$$

For user sequence $S_u = (i_1, \dots, i_t)$:

$$\mathbf{x}_k = \mathbf{z}_{i_k} + \mathbf{p}_k, \quad \mathbf{h}_u = \text{Transformer}(\mathbf{x}_1, \dots, \mathbf{x}_t)[t], \quad (13)$$

where only interest embeddings $\mathbf{z}_{i_k}$ are used as input, and $\mathbf{p}_k$ is the learnable positional embedding for the k -th position in the sequence. The transformer loss is the standard full softmax cross-entropy:

$$L_{\text{local}} = - \sum_{(u,t)} \log \frac{\exp(\mathbf{h}_u^\top \mathbf{z}_{i_{t+1}})}{\sum_{j \in \mathcal{I}} \exp(\mathbf{h}_u^\top \mathbf{z}_j)}. \quad (14)$$

Here, j indexes all candidate items in the item set $\mathcal{I}$.

After batch standardization of $\tilde{\mathbf{h}}_u, \tilde{\mathbf{z}}_u, \tilde{\mathbf{v}}_u$:

$$\mathcal{L}_{\text{align}} = \sum_i (1 - C_{ii}^{hz})^2 + \lambda \sum_{i \neq j} (C_{ij}^{hz})^2 + \mu \sum_{i,j} (C_{ij}^{hv})^2, \quad (15)$$

where C^{hz} and C^{hv} are cross-correlation matrices between batch-normalized representations $(\tilde{\mathbf{h}}_u, \tilde{\mathbf{z}}_u)$ and $(\tilde{\mathbf{h}}_u, \tilde{\mathbf{v}}_u)$, respectively; the first two terms align $\mathbf{h}_u$ with interest $\mathbf{z}_u$ (Barlow Twins style), and the third enforces orthogonality to disinterest $\mathbf{v}_u$.

The total loss is

$$\mathcal{L} = L_{\text{local}} + w_{\text{global}} L_{\text{global}} + w_{\text{align}} \mathcal{L}_{\text{align}}. \quad (16)$$

Training proceeds in two phases: (1) Warm-up — train only Pone-GNN using L_{global} ; (2) Joint — end-to-end optimization of $\mathcal{L}$.

B. CREATE-Uni: Hypergraph Variant

Let U be the set of users, I the set of items and $V = U \cup I$. Each observed interaction is a pair (u, i, τ) where $u \in U$, $i \in I$ and $\tau \in \mathbb{R}$ is the interaction timestamp.

Fix a session length $t > 0$ (hours). For every user u we partition u 's timeline into s_u contiguous sessions of length t . Denote the r -th session time interval for user u by

$$T_{u,r} = [\tau_0^{(u)} + (r-1)t, \tau_0^{(u)} + rt], \quad r = 1, \dots, s_u$$

where $\tau_0^{(u)}$ is a chosen reference time for u (e.g. first interaction or a global anchor). The set of items that u interacted with during session r is

$$H_{u,r} := \{i \in I \mid \exists \tau \in T_{u,r} \text{ s.t. } (u, i, \tau) \text{ observed}\}.$$

Define the session hyperedge (for user u , session r) as

$$e_{u,r} := \{u\} \cup H_{u,r}.$$

The hypergraph is then

$$\mathcal{H} = (V, \mathcal{E}), \quad \mathcal{E} := \{e_{u,r} \mid u \in U, r = 1, \dots, s_u, H_{u,r} \neq \emptyset\}. \quad 1) \text{ Setup and Notation: For every user } u, \text{ the item universe } \mathcal{I} \text{ partitions into three disjoint subsets:}$$

We apply the two-stage UniGNN aggregation:

$$h_e = \phi_1(\{x_j\}_{j \in e}), \quad \tilde{x}_v = \phi_2(x_v, \{h_e\}_{e \in E_v}), \quad (17)$$

where ϕ_1 is mean aggregation and ϕ_2 is instantiated as UniGCN, UniGAT, or UniGIN. Final embeddings are layer-averaged:

$$\mathbf{g}_v = \frac{1}{K} \sum_{l=0}^{K-1} \mathbf{x}_v^{(l+1)} \quad \forall v \in \mathcal{V}. \quad (18)$$

We adopt pairwise ranking (BPR) on the hypergraph user-item scores:

$$L_{\text{global}} = - \sum_u \sum_{i \in H_u^+} \sum_{j \notin H_u^+} \ln \sigma(\mathbf{g}_u^\top \mathbf{g}_i - \mathbf{g}_u^\top \mathbf{g}_j). \quad (19)$$

For sequence $S_u = (i_1, \dots, i_t)$:

$$\mathbf{x}_k = \mathbf{g}_{i_k} + \mathbf{p}_k, \quad \mathbf{h}_u = \text{Transformer}(\mathbf{x}_1, \dots, \mathbf{x}_t)[t], \quad (20)$$

where $\mathbf{p}_k$ denotes the learnable positional embedding at position k . with loss

$$L_{\text{local}} = - \sum_{(u,t)} \log \frac{\exp(\mathbf{h}_u^\top \mathbf{g}_{i_{t+1}})}{\sum_j \exp(\mathbf{h}_u^\top \mathbf{g}_j)}. \quad (21)$$

Here, j indexes all candidate items in $\mathcal{I}$.

Standard Barlow Twins between the sequential state $\mathbf{h}_u$ and hypergraph user embedding $\mathbf{g}_u$:

$$\mathcal{L}_{\text{align}} = \sum_i (1 - C_{ii}^{hg})^2 + \lambda \sum_{i \neq j} (C_{ij}^{hg})^2. \quad (22)$$

Here, C^{hg} is the cross-correlation matrix between the batch-normalized representations $\tilde{\mathbf{h}}_u$ and $\tilde{\mathbf{g}}_u$.

Training follows the same two-phase schedule as CREATE-Pone.

At inference, both variants use only the sequential encoder with the enriched item embeddings ($\mathbf{z}_i$ for CREATE-Pone, $\mathbf{g}_i$ for CREATE-Uni). For a user representation $\mathbf{h}_u$, prediction scores are computed by inner product with enriched item embeddings, i.e., $\hat{y}_{u,i} = \mathbf{h}_u^\top \mathbf{e}_i$, where $\mathbf{e}_i = \mathbf{z}_i$ for CREATE-Pone and $\mathbf{e}_i = \mathbf{g}_i$ for CREATE-Uni.

IV. THEORETICAL ANALYSIS

We organise the analysis into two parts. Section IV-A justifies CREATE-Pone by proving that unsigned LightGCN suffers from two concrete deficiencies — *exposure bias* and *polarity blindness* — and that our signed dual-branch propagation with orthogonality alignment provably eliminates both. Section IV-B justifies CREATE-Uni by invoking the established 1-WL / 1-GWL expressivity hierarchy [3], [15] and constructing an explicit user pair that separates our setting.

Throughout, $\|\cdot\|_F$ is the Frobenius norm, σ is the sigmoid, and $\mathbb{E}[\cdot]$ is expectation over the training distribution.

A. CREATE-Pone: Signed Modeling Advantages

1) *Setup and Notation:* For every user u , the item universe $\mathcal{I}$ partitions into three disjoint subsets:

$$\mathcal{I} = \underbrace{\mathcal{O}_u}_{\text{observed positives}} \cup \underbrace{\mathcal{D}_u}_{\text{observed negatives}} \cup \underbrace{\mathcal{S}_u}_{\text{never observed}}, \quad (23)$$

where $\mathcal{O}_u$ denotes items the user has explicitly liked (e.g. ratings ≥ 4), $\mathcal{D}_u$ denotes items the user has explicitly disliked (e.g. ratings ≤ 3), and $\mathcal{S}_u$ denotes items the user has never interacted with at all. All three sets are disjoint by construction.

Standard LightGCN observes $\mathcal{O}_u \cup \mathcal{D}_u$ as edges in the interaction graph, treating both liked and disliked interactions as equally positive signal. BPR then samples negatives only from $\mathcal{S}_u$ — items the user has never encountered. This means the model is corrupted in two simultaneous ways: disliked items from $\mathcal{D}_u$ pollute the user embedding as if they were positive (Limitation 2), and the model never receives any supervision signal from $\mathcal{D}_u$ as the negative evidence it actually represents (Limitation 1). We analyse each in turn.

2) *Limitation 1 — Exposure Bias in Unsigned BPR*: Since LightGCN folds $\mathcal{D}_u$ into the positive graph, BPR samples negatives exclusively from $\mathcal{S}_u$ — items the user has simply never been exposed to. However, $\mathcal{S}_u$ contains no genuine dislike signal: these are items the user may well enjoy but has never encountered. In practice $|\mathcal{S}_u| \gg |\mathcal{D}_u|$ for most users, so the model is trained to rank observed items above a vast pool of potentially enjoyable unseen items, receiving noisy and misleading supervision throughout.

This is the *exposure bias*: the true negative evidence in $\mathcal{D}_u$ is never used as such, while $\mathcal{S}_u$ — which carries no genuine negative signal — is treated as the sole source of negative supervision.

Resolution in CREATE-Pone. The dual feedback loss $\mathcal{L}_{\text{DF}}$ (Eq. 9) draws negatives exclusively from $\mathcal{D}_u \equiv E_n$ — items the user has explicitly disliked. Items from $\mathcal{S}_u$ are never used as negative evidence, ensuring that every negative training signal is semantically meaningful.

3) *Limitation 2 — Polarity Blindness in LightGCN*: Because LightGCN places all edges from $\mathcal{O}_u \cup \mathcal{D}_u$ into a single unsigned graph, it has no awareness that $\mathcal{D}_u$ represents negative feedback. Every item a user has rated — whether liked or disliked — is treated as a positive neighbour and contributes to the user embedding via message passing in exactly the same way.

This is the *polarity blindness* of unsigned graphs. The user embedding becomes an indiscriminate mixture of interest and disinterest signals, with no mechanism to separate them. As a result, items in $\mathcal{D}_u$ actively pull the user representation toward themselves during aggregation — the same direction as liked items. Any candidate item j similar to a disliked item $a \in \mathcal{D}_u$ will therefore receive an *inflated* recommendation score via the inner product $\hat{y}_{uj} = e_u^\top e_j$, since a 's embedding has been folded into e_u as if it were a preference. The more disliked items a user has interacted with, the more corrupted their embedding becomes.

Resolution in CREATE-Pone. The dual-branch construction propagates interest embeddings z_u solely on G_p (built from $\mathcal{O}_u$) and disinterest embeddings v_u solely on G_n (built from $\mathcal{D}_u$). Since G_p and G_n share no edges, the two branches are completely isolated — items from $\mathcal{D}_u$ can never corrupt z_u , and the interest signal is preserved cleanly.

4) *Theorem 1 — Orthogonality Loss Disentangles h_u from v_u* : **Statement.** Minimising the orthogonality term $\mu \|C^{hv}\|_F^2 \rightarrow 0$ eliminates all linear dependence between the sequential state h_u and the disinterest embedding v_u .

Proof. After batch normalisation, each representation has zero mean. C_{ij}^{hv} is therefore exactly the sample cross-correlation between the i -th dimension of h_u and the j -th dimension of v_u . Driving $\|C^{hv}\|_F^2 \rightarrow 0$ forces every entry of C^{hv} to zero, meaning h_u and v_u are *uncorrelated* across all dimensions. Uncorrelated representations have no linear relationship: no linear function of v_u can predict any dimension of h_u better than a constant.

This matters directly for recommendation, since prediction scores are computed as the inner product $\hat{y}_{ui} = h_u^\top z_i$. An

h_u that is uncorrelated with v_u cannot inadvertently encode disinterest signals from $\mathcal{D}_u$ into its recommendation scores.

Under the additional assumption that (h_u, v_u) are jointly Gaussian, zero correlation further implies full statistical independence $I(h_u; v_u) = 0$, though the linear guarantee above is sufficient in practice. $\square$

Pull–push duality. The full alignment loss (Eq. 15) has two complementary roles:

- **Pull** — $\|C^{hz} - I\|_F^2 \rightarrow 0$ aligns h_u toward the interest subspace of z_u , so the sequential encoder inherits collaborative signals from $\mathcal{O}_u$ (Barlow Twins [9]).
- **Push** — $\|C^{hv}\|_F^2 \rightarrow 0$ repels h_u away from the disinterest subspace of v_u , preventing signals from $\mathcal{D}_u$ from leaking into sequential recommendations.

No such push exists in the original unsigned CREATE. Since it operates on a single unsigned graph over $\mathcal{O}_u \cup \mathcal{D}_u$, the sequential encoder can silently absorb aversion signals through the shared item embedding layer with no loss term to prevent it. CREATE-Pone's alignment is therefore strictly richer.

B. CREATE-Uni: Hypergraph Expressivity

1) *Established Results We Rely On*: We use three results from the literature without reproving them:

- **GNNs $\leq$ 1-WL** [15]: Any message-passing GNN (including LightGCN) is at most as powerful as the 1-dimensional Weisfeiler-Leman (1-WL) graph isomorphism test. Two nodes that 1-WL cannot distinguish will receive identical embeddings from any such GNN.
- **UniGNN $\equiv$ 1-GWL** [3]: When the aggregation functions ϕ_1 (node-to-hyperedge) and ϕ_2 (hyperedge-to-node) are injective on multisets, UniGNN is equivalent in power to the 1-dimensional *Generalized* Weisfeiler-Leman (1-GWL) test on hypergraphs.
- **1-GWL $\geq$ 1-WL** [16]: 1-GWL is strictly more powerful than 1-WL on structures containing multi-way relationships, such as hyperedges of size ≥ 2 , which pairwise graphs cannot natively represent.

What remains to be shown is that the specific user-modeling scenario in CREATE-Uni actually *instantiates* the strict gap — i.e., that there is a concrete, interpretable user pair that LightGCN conflates but UniGNN separates.

2) *Proposition 1 — LightGCN Conflates Users with Identical Item Neighbourhoods*: **Claim.** If two users u_1, u_2 have $N(u_1) = N(u_2)$ (the same set of interacted items) in the bipartite graph, LightGCN assigns them identical embeddings at every layer.

Proof. By induction on layer k .

Base case ($k = 0$): Users are initialised from the same distribution (standard practice); assume WLOG $e_{u_1}^{(0)} = e_{u_2}^{(0)}$.

Inductive step: Assume $e_{u_1}^{(k)} = e_{u_2}^{(k)}$ and every item $i \in N(u_1) = N(u_2)$ has the same embedding $e_i^{(k)}$ (same item

node, same embedding). Then

$$\begin{aligned}
e_{u_1}^{(k+1)} &= \sum_{i \in N(u_1)} \frac{e_i^{(k)}}{\sqrt{|N(u_1)||N(i)|}} \\
&= \sum_{i \in N(u_2)} \frac{e_i^{(k)}}{\sqrt{|N(u_2)||N(i)|}} \\
&= e_{u_2}^{(k+1)}.
\end{aligned} \tag{24}$$

The layer-averaged final embeddings are therefore identical. Formally, this follows from $N(u_1) = N(u_2)$ and the fact that 1-WL cannot distinguish nodes with identical neighbourhood multisets [15].

3) *Proposition 2 — UniGNN Separates Users that Share Item Neighbourhoods but Have Different Session Structure:*

Claim. There exist users u_1, u_2 with $N(u_1) = N(u_2)$ (so LightGCN conflates them) whose session structure differs, and UniGNN with injective aggregation assigns them distinct embeddings.

Construction. Let $\mathcal{I} = \{a, b, c, d\}$ with distinct initial embeddings.

- u_1 has **one session** $\{a, b, c, d\}$ — a single hyperedge $e_1 = \{a, b, c, d\}$.
- u_2 has **two sessions** $\{a, b\}$ and $\{c, d\}$ — hyperedges $e_2 = \{a, b\}$ and $e_3 = \{c, d\}$.

Both users have the same bipartite neighbourhood $N(u_1) = N(u_2) = \{a, b, c, d\}$.

LightGCN cannot separate them. By Proposition IV-B2, $e_{u_1}^{(k)} = e_{u_2}^{(k)}$ for all k .

UniGNN separates them. Using sum aggregation for ϕ_1 :

$$\begin{aligned}
h_{e_1} &= e_a + e_b + e_c + e_d \quad (\text{one hyperedge, size 4}), \\
h_{e_2} &= e_a + e_b, \quad h_{e_3} = e_c + e_d \quad (\text{two hyperedges, size 2 each}).
\end{aligned}$$

Now ϕ_2 aggregates the *multiset of incident hyperedge embeddings*:

$$\begin{aligned}
\tilde{e}_{u_1}^{(1)} &= \phi_2\left(e_{u_1}^{(0)}, \underbrace{\{\{h_{e_1}\}\}}_{1 \text{ element}}\right), \\
\tilde{e}_{u_2}^{(1)} &= \phi_2\left(e_{u_2}^{(0)}, \underbrace{\{\{h_{e_2}, h_{e_3}\}\}}_{2 \text{ elements}}\right).
\end{aligned}$$

The second arguments are multisets of *different cardinality* (1 vs. 2) and different elements. ϕ_2 maps distinct multisets to distinct outputs:

$$\tilde{e}_{u_1}^{(1)} \neq \tilde{e}_{u_2}^{(1)}.$$

UniGNN successfully distinguishes the two users.

Semantic interpretation. u_1 bought $\{a, b, c, d\}$ *together* (e.g., a complete skincare routine in one basket) — the co-purchase group is a meaningful collaborative signal. u_2 bought $\{a, b\}$ and $\{c, d\}$ on separate occasions — each session carries its own coherence signal. LightGCN collapses these into the same representation; UniGNN encodes the structural difference, yielding richer item embeddings g_i for the sequential encoder to align with.

4) *Corollary — CREATE-Uni Provides Richer Alignment Targets than CREATE: Statement.* The Barlow Twins alignment loss $\mathcal{L}_{\text{align}}$ (Eq. 23) conveys strictly more discriminative gradient signal to the sequential encoder under CREATE-Uni than under the original CREATE.

Proof sketch. By Proposition IV-B3, u_1 and u_2 receive different hypergraph embeddings: $g_{u_1} \neq g_{u_2}$. In a batch containing both users, the batch-normalised targets $\tilde{g}_{u_1} \neq \tilde{g}_{u_2}$ contribute different columns to the cross-correlation matrix C^{hg} , so the alignment loss generates informative gradients that push h_{u_1} and h_{u_2} apart.

Under LightGCN, $z_{u_1} = z_{u_2}$ (Proposition IV-B2), so both users produce identical alignment targets. The sequential encoder receives no gradient from $\mathcal{L}_{\text{align}}$ to differentiate h_{u_1} from h_{u_2} , even though their behavioral patterns differ. The mutual information transferred via alignment is therefore strictly greater in CREATE-Uni.

V. DATASETS

We evaluate our method on three widely used recommendation datasets: **Amazon Beauty (2014)** [17], **Amazon Office Products (2014)** [18], and **MovieLens-1M** [19]. All datasets can be viewed as bipartite graphs $\mathcal{G} = (\mathcal{U}, \mathcal{I}, \mathcal{E})$, where $\mathcal{U}$ denotes users, $\mathcal{I}$ denotes items, and $\mathcal{E}$ denotes user-item interactions. Hyperedges are formulated as described in the CREATE-Uni variant.

Amazon Beauty (2014): This dataset is a subset of the Amazon Review dataset containing user reviews and metadata for beauty products. User nodes correspond to reviewers and item nodes correspond to beauty-related products. Edges represent review interactions between users and products. The dataset includes review ratings, review text, timestamps, helpfulness votes, and product metadata such as descriptions, categories, price, brand, and image features.

Amazon Office Products (2014): This dataset contains user reviews and metadata for office-related products from Amazon. User nodes represent reviewers and item nodes correspond to office products. Edges denote user review interactions. The dataset includes ratings, review text, timestamps, and product metadata including category information, descriptions, brand, price, and image features.

MovieLens-1M: MovieLens-1M contains 1,000,209 anonymous movie ratings from 6,040 users on approximately 3,900 movies. User nodes correspond to MovieLens users and item nodes correspond to movies. Edges represent user rating interactions with movies. The dataset includes explicit rating values and timestamps. Additional user demographic information such as gender, age, occupation, and zip code is also available, along with movie genre metadata.

TABLE I: Dataset Statistics: Core Graph Properties

Dataset	#Users	#Items	#Interactions	Sparsity
Amazon Beauty	22,363	12,101	528,417	0.99805
ML-1M	6,040	3,706	2,982,507	0.86676
Amazon Office	4,905	2,420	145,059	0.98778

TABLE II: Dataset Statistics: Sequence Length Summary

Dataset	Seq Mean	Seq Med	Seq Min
Amazon Beauty	23.63	15.0	12.0
ML-1M	493.79	285.0	57.0
Amazon Office	29.57	18.0	12.0

TABLE III: Dataset Statistics: Session Structure

Dataset	Avg Sess Size	Avg Sess/User	Session Len (s)
Amazon Beauty	5.66	4.18	2592000
ML-1M	134.47	3.67	86400
Amazon Office	3.29	9.00	7776000

VI. BASELINES USED

We compare **CREATE-Uni** with four representative baselines: **SASRec** [4], **CREATE** [1], **LightGCN** [8], and **UniGCN**. These baselines cover the three most relevant settings for our study: a purely sequential model (SASRec), the original cross-representation alignment framework (CREATE), and graph/hypergraph collaborative filtering models without sequential alignment (LightGCN and UniGCN).

This comparison lets us answer three questions directly: whether hypergraph alignment improves over a strong sequential recommender, whether it improves over the original CREATE framework, and whether the gains come from higher-order structure alone or from combining higher-order structure with sequential modeling. Together, these baselines provide a compact but informative reference for evaluating the contribution of CREATE-Uni.

VII. RESULTS

Table IV reports the main comparison between CREATE-Uni and the baselines. For CREATE-Uni, we now report the average performance across three random seeds together with the standard deviation. Across all three datasets, CREATE-Uni still achieves the best overall performance on the reported ranking metrics, with the most consistent gains appearing on **MovieLens-1M** and **Office Products**. In particular, the model improves mean NDCG@10 from 6.25 to 6.69 ± 0.15 on Office Products and from 10.34 to 11.21 ± 0.17 on MovieLens-1M, indicating that higher-order structural alignment improves the quality of the top-ranked recommendations.

The seed-wise mean and standard deviation are summarized in Table V. Across all three datasets, the variance remains small relative to the absolute metric values, which indicates that CREATE-Uni is stable across random initializations. Beauty shows especially tight variation, while Office Products and MovieLens-1M exhibit slightly larger but still well-controlled spread. In addition, Table VI reports the relative improvement of the mean CREATE-Uni performance over the best non-CREATE-Uni baseline for each metric. The largest gains appear on MovieLens-1M, while Beauty shows smaller but still consistent improvements. Combined with the gains over SASRec and CREATE in Table IV, these statistics support our central hypothesis that higher-order hypergraph alignment yields improvements that are both effective and reproducible.

Another clear observation is that the graph-only baselines, especially LightGCN and UniGCN, perform substantially below the sequential and alignment-based models in these settings. This is consistent with the deployment motivation behind CREATE-style methods: graph encoders are useful for providing structural supervision during training, but relying on them alone is often insufficient for capturing the fine-grained temporal dynamics needed for next-item prediction. CREATE-Uni benefits precisely from combining these two views rather than replacing one with the other.

The training-dynamics plots in Figures 1–3 provide a complementary view of this behavior. Across datasets, the loss curves help verify that the local, global, and alignment objectives are jointly optimized in a stable manner, while the metric curves show that **NDCG@10**, **Recall@10**, and **Precision@10** improve together over training before stabilizing near convergence. This agreement across all three evaluation metrics indicates that the gains of CREATE-Uni are not limited to a single measure, but consistently improve both ranking quality and top- k retrieval behavior. These plots are especially useful for diagnosing whether performance gains are accompanied by healthy optimization dynamics rather than isolated endpoint improvements.

CREATE-Pone remains a conceptually meaningful extension of CREATE because signed modeling offers a principled way to separate user interest from disinterest. However, evaluating this variant thoroughly requires substantially more care in constructing reliable signed supervision, as well as greater computational cost due to the added optimization objectives and more involved training dynamics. Given the time and compute constraints of this project, we chose to focus our empirical analysis on **CREATE-Uni**, which was more practical to run extensively across datasets, seeds, and ablation settings. We therefore view CREATE-Pone as a promising direction for future work rather than the central focus of the present study.

VIII. EXPERIMENTS

A. Experimental Setup

We evaluate our models on three datasets for which we obtained stable runs with CREATE-Uni: **Amazon Beauty**, **Office Products**, and **MovieLens-1M**. Following standard sequential recommendation practice, each user’s interaction history is ordered chronologically, and the last interactions are reserved for validation and test evaluation. We report **NDCG@10**, **Recall@10**, and **Precision@10**, which together measure ranking quality, retrieval coverage, and top- k recommendation accuracy.

For all methods, the sequential branch uses the same backbone architecture and embedding dimensionality to ensure fair comparison. For CREATE and CREATE-Uni, the graph-side objective is optimized jointly with the sequential next-item prediction loss and the alignment loss. Hyperparameters such as learning rate, batch size, and alignment weight are tuned on the validation split, and the final model is selected based on validation NDCG@10.

TABLE IV: Test-set comparison of sequential and graph-based baselines with CREATE-Uni. All values are reported in percentage (%).

Dataset	Metric	SASRec	CREATE	UniGCN	LightGCN	CREATE-Uni
Beauty	NDCG@10	3.98	4.53	1.40	1.18	4.86 $\pm$ 0.03
	Recall@10	8.13	9.44	3.06	2.55	9.48 $\pm$ 0.05
	Precision@10	0.01	0.94	0.31	0.25	0.95 $\pm$ 0.01
Office Products	NDCG@10	5.53	6.25	1.07	1.41	6.69 $\pm$ 0.15
	Recall@10	11.68	13.09	2.71	3.17	13.48 $\pm$ 0.23
	Precision@10	1.17	1.31	0.27	0.32	1.35 $\pm$ 0.02
MovieLens-1M	NDCG@10	10.34	9.65	1.49	1.49	11.21 $\pm$ 0.17
	Recall@10	20.51	20.31	3.26	3.36	22.64 $\pm$ 0.18
	Precision@10	2.05	2.03	0.33	0.34	2.26 $\pm$ 0.02

TABLE V: Seed-wise mean and standard deviation of CREATE-Uni across datasets. All values are reported in percentage (%).

Dataset	NDCG@10	Recall@10	Precision@10
Beauty	4.86 $\pm$ 0.03	9.48 $\pm$ 0.05	0.95 $\pm$ 0.01
Office Products	6.69 $\pm$ 0.15	13.48 $\pm$ 0.23	1.35 $\pm$ 0.02
MovieLens-1M	11.21 $\pm$ 0.17	22.64 $\pm$ 0.18	2.26 $\pm$ 0.02

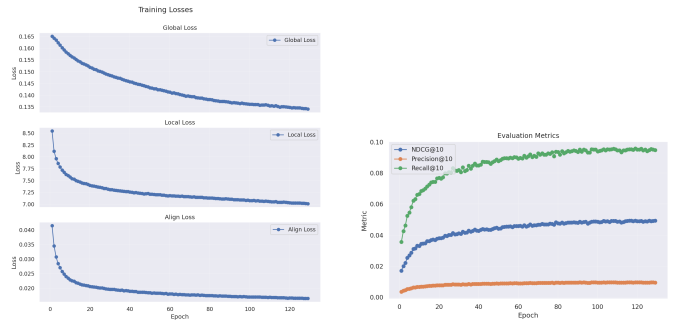
TABLE VI: Relative improvement of mean CREATE-Uni performance over the best non-CREATE-Uni baseline for each metric.

Dataset	Metric	Improvement (%)
Beauty	NDCG@10	+7.21
Beauty	Recall@10	+0.46
Beauty	Precision@10	+0.71
Office Products	NDCG@10	+7.04
Office Products	Recall@10	+2.95
Office Products	Precision@10	+3.05
MovieLens-1M	NDCG@10	+8.38
MovieLens-1M	Recall@10	+10.37
MovieLens-1M	Precision@10	+10.41

In addition to the main comparison, we design three targeted experiments to understand the behavior of CREATE-Uni beyond raw recommendation accuracy.

B. Training Dynamics

To better understand how CREATE-Uni trains in practice, we also inspect two diagnostic plots for each dataset: (i) the evolution of the three optimization terms—global loss, local loss, and alignment loss—over epochs, and (ii) the change in the ranking metrics—NDCG@10, Recall@10, and Precision@10—over epochs. These figures help connect the final test-set numbers to the underlying optimization process.



(a) Beauty: global, local, and alignment losses over epochs.

(b) Beauty: NDCG@10, Recall@10, and Precision@10 over epochs.

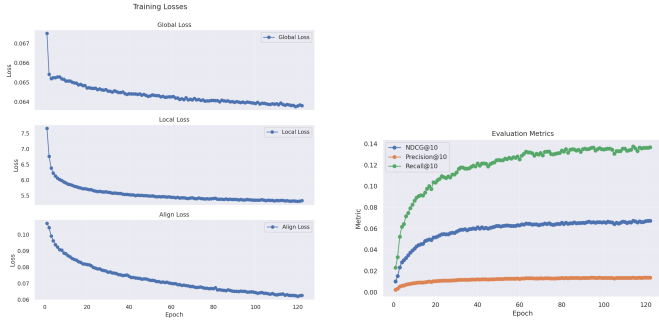
Fig. 1: Training dynamics of CREATE-Uni on the Beauty dataset.

In particular, they allow us to verify whether the three losses decrease in a coordinated manner and whether the evaluation metrics improve steadily before reaching a stable plateau.

C. Oversmoothing Analysis

Hypergraph message passing can improve structural expressiveness, but increasing the number of propagation layers may eventually lead to oversmoothing, where node or hyperedge representations become less distinguishable. To study this effect, we vary the number of graph propagation layers from 1 to 4 and track **NDCG@10**, **Recall@10**, and **Precision@10**. Considering all three metrics is useful because oversmoothing may affect ranking quality and top- k retrieval in slightly different ways: NDCG captures the quality of the ranking order, while Recall and Precision reflect retrieval coverage and accuracy.

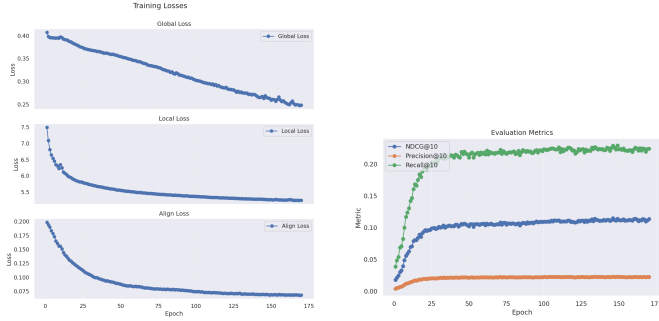
We expect a similar non-monotonic trend across these metrics. Moving from very shallow propagation to a moderate number of layers should improve performance by allowing CREATE-Uni to incorporate richer higher-order context, but



(a) Office Products: global, local, and alignment losses over epochs.

(b) Office Products: NDCG@10, Recall@10, and Precision@10 over epochs.

Fig. 2: Training dynamics of CREATE-Uni on the Office Products dataset.



(a) MovieLens-1M: global, local, and alignment losses over epochs.

(b) MovieLens-1M: NDCG@10, Recall@10, and Precision@10 over epochs.

Fig. 3: Training dynamics of CREATE-Uni on the MovieLens-1M dataset.

further increasing depth is likely to wash out discriminative information and reduce all three scores. The oversmoothing analysis therefore helps identify the depth regime in which CREATE-Uni best balances expressive structural aggregation against representation collapse.

Figures 4, 5, and 6 present the oversmoothing analysis on the MovieLens-1M dataset for both CREATE and CREATE-Uni. Across all three metrics, we observe a clear non-monotonic trend with respect to graph depth. Performance improves when increasing the number of propagation layers from 1 to 3, indicating that deeper propagation enables the model to capture richer higher-order collaborative signals and long-range user-item dependencies.

The best performance for both models is consistently achieved at 3 graph layers. In particular, CREATE-Uni reaches its peak values of approximately 11.35 NDCG@10, 0.227 Recall@10, and 0.0227 Precision@10 at this depth. This suggests that moderate propagation depth provides an effective

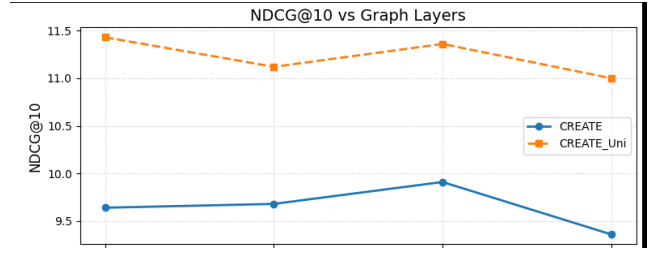


Fig. 4: Oversmoothing analysis on MovieLens-1M: NDCG@10 versus number of graph layers for CREATE and CREATE-Uni.

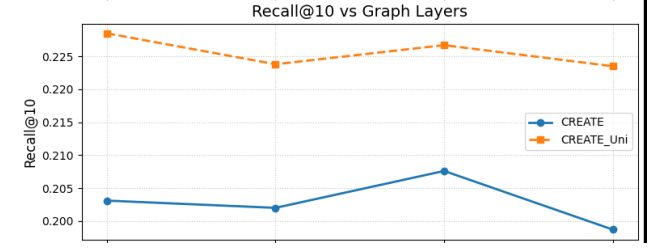


Fig. 5: Oversmoothing analysis on MovieLens-1M: Recall@10 versus number of graph layers for CREATE and CREATE-Uni.

balance between structural expressiveness and representation discrimination.

However, when the depth is further increased to 4 layers, all evaluation metrics decline for both CREATE and CREATE-Uni. The drop is especially noticeable for the baseline CREATE model, where NDCG@10, Recall@10, and Precision@10 decrease substantially. This behavior is consistent with the classical oversmoothing phenomenon in graph-based learning, where repeated propagation causes node representations to become increasingly similar, thereby reducing the model’s ability to distinguish between relevant and irrelevant items.

Compared with CREATE, CREATE-Uni exhibits significantly greater robustness to increasing graph depth. Although CREATE-Uni also experiences a slight decline at 4 layers, the degradation is much smaller across all metrics. This indicates that the proposed unified hypergraph representation and multi-view structural modeling help preserve discriminative information even under deeper propagation. Overall, the results suggest that CREATE-Uni achieves a better trade-off between higher-order information aggregation and resistance to representation collapse.

D. Robustness to Edge Removal

Real-world interaction graphs are often noisy or incomplete. To evaluate robustness, we randomly remove **10%–50%** of training edges and re-train/evaluate both models. This stress test measures how well CREATE-Uni preserves ranking quality as the interaction structure becomes progressively sparser.

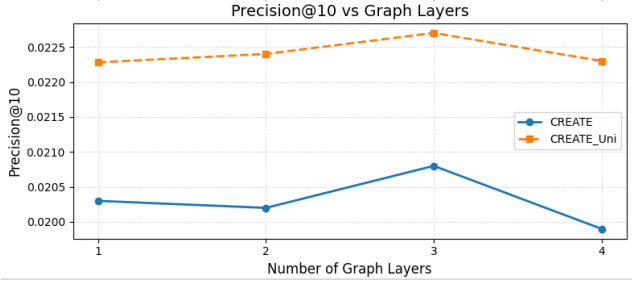


Fig. 6: Oversmoothing analysis on MovieLens-1M: Precision@10 versus number of graph layers for CREATE and CREATE-Uni.

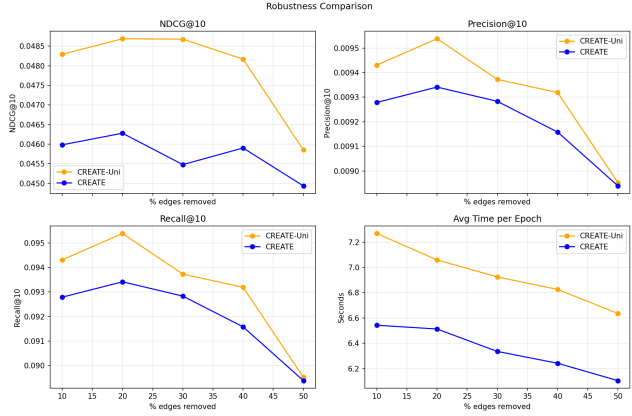


Fig. 7: Robustness Analysis on Amazon Beauty: NDCG@10, Recall@10, Precision@10, and average epoch time versus random edge removal.

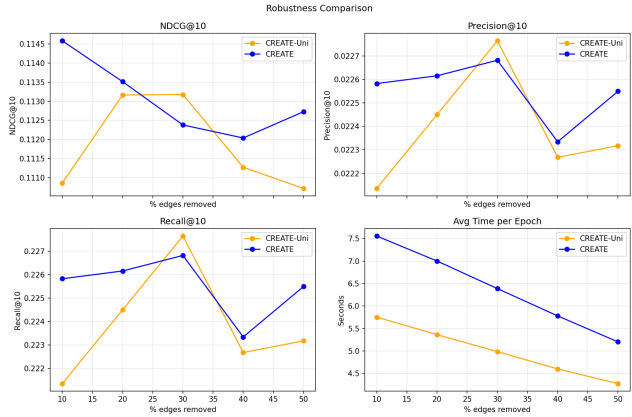


Fig. 8: Robustness Analysis on MovieLens-1M: NDCG@10, Recall@10, Precision@10, and average epoch time versus random edge removal.

IX. CONCLUSION

In this work, we extended the CREATE framework in two directions designed to overcome the limitations of unsigned bipartite alignment: signed modeling through CREATE-Pone and higher-order hypergraph modeling through CREATE-Uni.

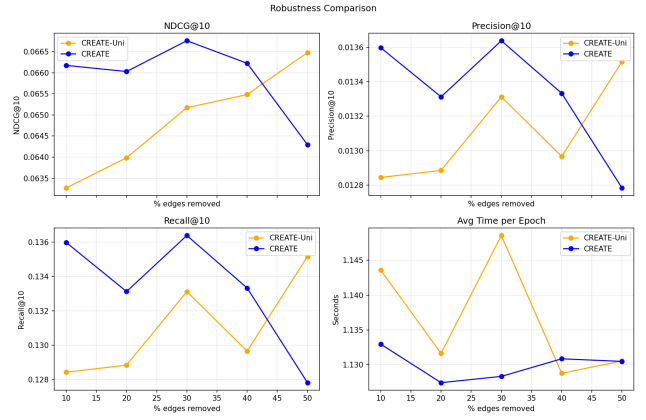


Fig. 9: Robustness Analysis on Office Products: NDCG@10, Recall@10, Precision@10, and average epoch time versus random edge removal.

Although the signed extension was conceptually motivated by the need to disentangle interest from disinterest, our empirical findings show that it is the hypergraph variant that delivers the clearest practical benefit. Across multiple datasets, CREATE-Uni consistently outperforms strong sequential, graph, and alignment-based baselines, demonstrating that higher-order collaborative structure provides richer supervision for sequential recommendation than pairwise graph propagation alone.

Beyond the main accuracy gains, our experimental design highlights three desirable properties of CREATE-Uni: it can be analyzed for oversmoothing across propagation depth, it remains meaningful under edge sparsification, and its results can be reported reliably across random seeds using mean $\pm$ standard deviation. These properties make CREATE-Uni a promising direction for recommendation systems that require both expressive structural modeling and efficient sequential inference.

As future work, the most natural next step is to improve the handling of negative and ambiguous feedback so that the benefits of signed modeling can be realized more consistently. Another promising direction is to construct richer hyperedges from sessions, bundles, or semantic item groups, and to study whether adaptive hypergraph formation can further improve robustness and personalization. Overall, our study suggests that the most effective way to extend CREATE is not merely to add more supervision, but to provide better structural targets for alignment.

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